

Effects of soy as tofu vs meat on lipoprotein concentrations.

OBJECTIVES: To investigate the effect of replacing lean meat with a soy product, tofu, on serum lipoprotein concentrations. **STUDY AND DESIGN:** Randomized cross-over dietary intervention study. **SUBJECTS:** Forty-two free-living healthy males aged 35-62 y completed the dietary intervention. Three additional subjects were non-compliant and excluded prior to analysis. **INTERVENTIONS:** A diet containing lean meat (150 g/d) was compared with one with 290 g/d tofu in an isocaloric and isoprotein substitution. Both diet periods were 1 month, and fat intake was carefully controlled. **RESULTS:** Seven-day diet records showed the two diets were similar in energy, macronutrients and fibre. Total cholesterol (mean difference 0.23 mmol/l, 95% CI 0.02, 0.43; $P=0.03$) and triglycerides (mean difference 0.15 mmol/l, 95% CI 0.02, 0.31; $P=0.017$) were significantly lower on the tofu diet than the lean meat diet. However, HDL-C was also significantly lower on the tofu diet (mean difference 0.08 mmol/l, 95% CI 0.02, 0.14; $P=0.01$) although the LDL-C:HDL-C ratio was similar. **CONCLUSION:** The effect on HDL-C and the small LDL-C reduction differ from some other studies, where fat was often less controlled, and the comparison was of soy as textured protein or soymilk against casein. This suggests a differential effect of the various proteins compared to the soy may influence the findings. In practice, the replacement of meat with tofu would usually be associated with a decrease in saturated fat and an increase in polyunsaturated fat and this should enhance any small benefits due to the soy protein. **SPONSOR:** Deakin University with some contribution from a Commonwealth Department of Veterans Affairs research grant. *European Journal of Clinical Nutrition* (2000) 54, 14-19